

MiniRAG: Towards Extremely Simple Retrieval-Augmented Generation

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자연어처리팀 - 이예서(발표자), 정현석, 박민건, 황예진, 박주혜, 김보성, 임다예슬, 김유진, 이웅기, 김기환, 조해창

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- Framework
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Introduction

SLM 사용시 발생하는 기존 RAG의 문제점

- LLM 의존적 설계
- SLM에서 성능 급격 저하
- 복잡한 파이프라인

핵심 연구 질문

"Small Language Model(SLM)에서도 효과적으로 작동하는 RAG 시스템을 어떻게 설계할 수 있을까?"

3B-4B 파라미터 규모의 경량 모델

(Phi-3.5-mini, Qwen2.5-3B, MiniCPM3-4B)

Introduction

Indexing 한계

SLM은 Entity 추출 시 LLM보다 품질이 낮은 설명을 생성

Answering 한계

SLM은 대량의 Context에서 관련 정보를 찾는 데 어려움

Limitation of Indexing

Original Text:

LiHua: Hey Adam! Hope you're doing well. Just wanted to ask for the Wi-Fi password when you get a chance. Also, are there any house rules I should know about? Thanks!

AdamSmith: Sure! The Wi-Fi password is "Family123". As for house rules, please keep noise to a minimum during late hours and take good care of the property. Let me know if you have any other questions!

LiHua: Got it, thanks! I'll make sure to keep it down at night. Just one more thing, is it okay to have friends over occasionally?

AdamSmith: You're welcome! Yes, having friends over occasionally is fine, just try to keep the gatherings reasonable. Enjoy!

LiHua: Awesome, appreciate it! Looking forward to settling in.

gpt-4o-mini response:

Entity: <HOUSE RULES>

Description: <House Rules refer to the guidelines provided by Adam Smith regarding noise and care of the property, aimed at maintaining a peaceful living environment.>

Phi-3.5-mini response:

Entity: <HOUSE RULES>

Description: <House rules are guidelines provided by AdamSmith for LiHua to follow while living in the house.>

Limitation of Answering

Input Prompt:

You are an assistant designed to answer questions. Please respond to the user's query based on the information provided by the user. If there is insufficient information, please reply with "I don't have enough information."

Here is the information:

Entities: <HOUSE RULES> ... and 49 other entities.

Relations: <LiHua, AdamSmith> ... and 49 other relationships.

Chunk: 10 chunks

Query: What are the House Rules mentioned by Adam Smith?

gpt-4o-mini response:

Based on <HOUSE RULES> and <LiHua, AdamSmith>, I know that the House Rules mentioned by Adam Smith is about...

Phi-3.5-mini response:

I'm sorry, I don't have enough information.

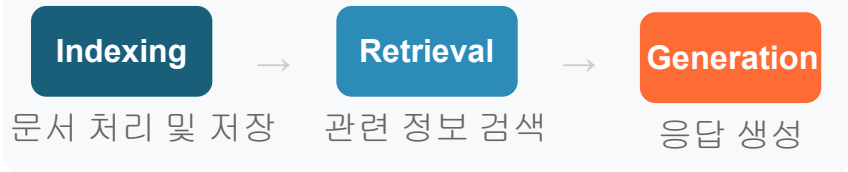
Figure 2: Compared to Large Language Models (LLMs), Small Language Models (SLMs) show significant limitations in both indexing and answering phases. **Left:** SLMs generate notably lower-quality descriptions than LLMs. **Right:** When processing identical inputs, SLMs struggle to locate relevant information in large contexts, while LLMs perform this task effectively.

Background

Retrieval-Augmented Generation (RAG)

외부 지식을 검색하여 LLM의 응답 생성을 보강하는 패러다임. 환각(Hallucination) 감소 및 최신 정보 반영 가능

RAG 파이프라인



RAG의 진화

Naive RAG

Vector 유사도 기반 검색

Advanced RAG

Pre/Post-retrieval 최적화

Graph RAG

Knowledge Graph 기반 구조화

Background

Vector RAG의 한계

- Multi-hop 추론 어려움
- Global Sensemaking 불가
- 복잡한 관계 파악 한계
- 분산된 정보 통합 실패

Graph RAG의 장점

- Entity-Relation 구조화
- 복잡한 관계 명시적 표현
- Multi-hop 추론 지원
- Global Query 처리 가능

비교 대상 Graph RAG 시스템

GraphRAG (Microsoft, 2024)

Hierarchical Community 기반 Global Sensemaking

LightRAG (HKU, 2024)

Dual-level Retrieval + Incremental Update

MiniRAG (HKU, 2025)

SLM 최적화 Heterogeneous Graph

Background

GraphRAG

복잡한 파이프라인: Entity 추출 → Community

감지 → Summary 생성

SLM 비호환: 고품질 Entity 설명 필요

높은 스토리지: Community Summary 저장

비용

LightRAG

Entity 품질 의존: LLM 수준 추출 필요

Keyword 매칭: SLM의 부정확한 키워드에 취약

Dedup 오버헤드: Entity 중복 제거 비용

Framework

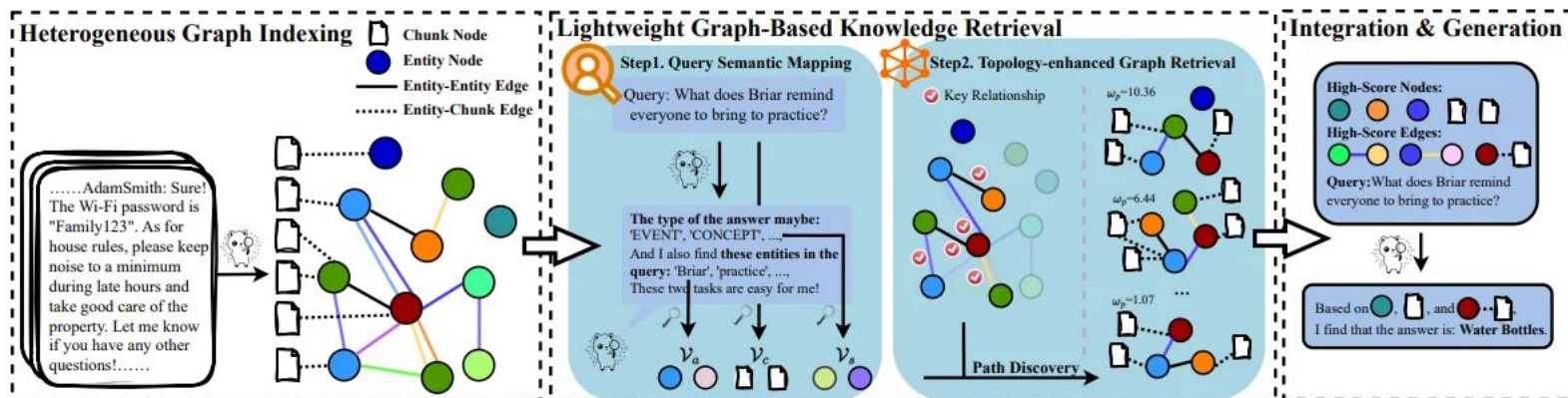


Figure 1: The MiniRAG employs a streamlined workflow built on the key components: heterogeneous graph indexing and lightweight graph-based knowledge retrieval. This architecture addresses the unique challenges faced by on-device RAG systems, optimizing for both efficiency and effectiveness.

1. Heterogeneous
Graph Indexing

2. Lightweight Retrieval

3. Integration & Generation

Framework: Heterogeneous Graph Indexing

그래프 정의

$$G = (\{\mathcal{V}_c, \mathcal{V}_e\}, \{\mathcal{E}_\alpha, (e_\beta, d_{e_\beta}) \in \mathcal{E}_\beta\})$$

노드 유형

Vc: Chunk 노드

문맥 정보를 담고 있는
원본 텍스트 조각

Ve: Entity 노드

텍스트에서 추출된 핵심
의미 요소 (이벤트, 장소
등)

엣지 유형

Eα: Entity-Entity

개체 간의 의미적 관계 및
계층 구조

Eβ: Entity-Chunk

개체와 그 개체가 포함된
텍스트 청크 간의 연결

핵심 설계 원칙

1. Semantic Indexing 최소화

SLM의 Entity 설명 생성 의존도 제거

2. Chunk를 First-class 노드로

원본 텍스트 직접 접근으로 정보 손실 방지

3. 이중 연결 구조

Entity 관계 + Chunk 출처 동시 추적

Framework: Heterogeneous Graph Indexing

목표

Query를 그래프의 Entry Points로 매핑하여 검색 범위를 축소

수행 단계

1. Answer Type 추론

예: EVENT, CONCEPT, LOCATION

2. Query Entity 추출

Query에서 핵심 Entity 식별

3. Entry Point 선택

그래프에서 매칭되는 노드 찾기

예시

Query:

"What does Briar remind everyone to bring to practice?"

Answer Type: EVENT, CONCEPT

Entities: Briar, practice

SLM도 수행 가능한 간단한 작업으로 설계

Framework: Topology-Enhanced Retrieval

핵심 아이디어

그래프 토폴로지를 활용하여 Query Entry Point에서 정답 후보까지의 추론 경로(Reasoning Path)를 탐색

Path Discovery

Entry Point에서 K-hop 이내 경로 탐색 → 경로 점수 기반으로 관련 Entity 및 Chunk 선별

핵심 설계

1. Semantic Matching 최소화

토폴로지 기반 관계 추론

2. Multi-hop 자연 지원

경로 기반 복합 추론

3. Key Relationship 식별

높은 점수의 엣지 → 핵심 관계

Framework: Topology-Enhanced Retrieval

점수 계산

$$\text{Path Score: } \omega_p(p \mid vq) = \omega_v(\hat{v}s \mid vq) \cdot (1 + \sum \text{count}(v, p) + \sum \omega_e(e))$$

$\omega_v(\hat{v}s \mid vq)$: 시작 노드와 쿼리 엔티티 간의 코사인 유사도

$\text{count}(v, p)$: 노드 v 가 경로 p 에 포함되면 1, 아니면 0

$\omega_e(e)$: 엣지의 관련성 점수

$$\text{Entity Score: } \omega_e(e) = \sum \omega_p(p \mid vq) \times \mathbb{1}(e \in p)$$

경로에 포함된 횟수와 경로 점수의 가중합

Framework: Integration & Generation

Context 구성 요소

High-Score Nodes 점수 기반 선별된 Entity + Chunk

High-Score Edges 핵심 관계 정보

Original Texts Chunk 노드의 원본 텍스트

Context 제한

Top-K: 5 / Max Tokens: 6000

Generation Flow

Query + Retrieved Context

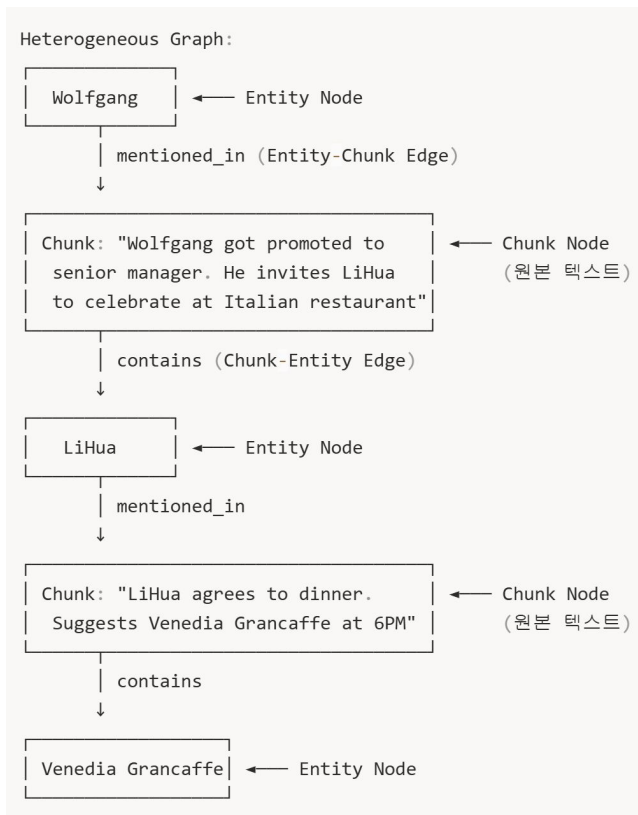


Small Language Model (SLM)



Final Answer

Framework: Example



Framework: MiniRAG vs GraphRAG, LightRAG

Graph Definition (Code View)



```
# GraphRAG
Graph = {
  "nodes": [Entity], # Entity만
  "edges": [Relationship],
  "structure": "Hierarchical Communities"
}
```



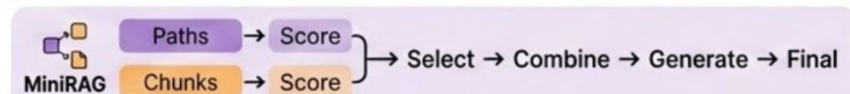
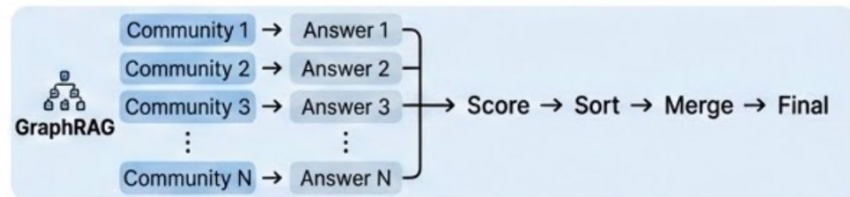
```
# LightRAG
Graph = {
  "nodes": [Entity], # Entity만 (GraphRAG과 동일)
  "edges": [Relationship + high-level keywords],
  "structure": "Flat Graph + Dual-level Index"
}
```



```
# MiniRAG
Graph = {
  "nodes": [Entity, Text Chunk], # 이중 구조
  "edges": [Entity-Entity, Entity-Chunk],
  "structure": "Heterogeneous Graph"
}
```

Key Characteristics & Process Flow

	 GraphRAG	 LightRAG	 MiniRAG
 Node Type	 Entity만	 Entity만	 Entity + Chunk
 Info Storage	 Community summary	 Entity/Relation description	 원본 Chunk 기반
 Structure	 계층적	 평면적 + 이중 index	 Heterogeneous



Evaluation

Table 1: Performance evaluation using accuracy (acc) and error (err) rates, measured as percentages (%). Higher accuracy and lower error rates indicate better RAG performance. Results compare various baseline methods against our MiniRAG across multiple datasets. Bold values indicate best performance, while “/” denotes cases where methods failed to generate effective responses.

LiHuaWorld	NaiveRAG		GraphRAG		LightRAG		MiniRAG	
	<i>acc</i> ↑	<i>err</i> ↓	<i>acc</i> ↑	<i>err</i> ↓	<i>acc</i> ↑	<i>err</i> ↓	<i>acc</i> ↑	<i>err</i> ↓
Phi-3.5-mini-instruct	41.22%	23.20%	/	/	39.81%	25.39%	53.29%	23.35%
GLM-Edge-1.5B-Chat	42.79%	24.76%	/	/	35.74%	25.86%	52.51%	25.71%
Qwen2.5-3B-Instruct	43.73%	24.14%	/	/	39.18%	28.68%	48.75%	26.02%
MiniCPM3-4B	43.42%	17.08%	/	/	35.42%	21.94%	51.25%	21.79%
gpt-4o-mini	46.55%	19.12%	35.27%	37.77%	56.90%	20.85%	54.08%	19.44%

MultiHop-RAG	NaiveRAG		GraphRAG		LightRAG		MiniRAG	
	<i>acc</i> ↑	<i>err</i> ↓	<i>acc</i> ↑	<i>err</i> ↓	<i>acc</i> ↑	<i>err</i> ↓	<i>acc</i> ↑	<i>err</i> ↓
Phi-3.5-mini-instruct	42.72%	31.34%	/	/	27.03%	11.78%	49.96%	28.44%
GLM-Edge-1.5B-Chat	44.44%	24.26%	/	/	/	/	51.41%	23.44%
Qwen2.5-3B-Instruct	39.48%	31.69%	/	/	21.91%	13.73%	48.55%	33.10%
MiniCPM3-4B	39.24%	31.42%	/	/	19.48%	10.41%	47.77%	26.88%
gpt-4o-mini	53.60%	27.19%	60.92%	16.86%	64.91%	19.37%	68.43%	19.41%

Evaluation: Accuracy vs Storage Efficiency

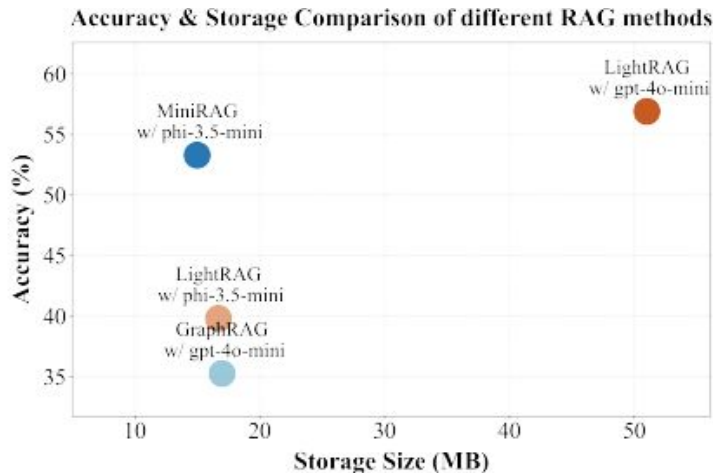


Figure 3: Accuracy vs. Storage Efficiency: Comparative analysis of three RAG systems - MiniRAG, LightRAG, and GraphRAG.

Storage & Accuracy 비교

- Community Summary 제거로 대폭 저장 공간 절감
- 동일 모델 비교시 큰 폭의 Accuracy 차이
- gpt-4o-mini 대비 성능 하락폭 최소화
- On-device 환경에 이상적: 제한된 스토리지에서 높은 성능

Evaluation: Ablation Study

- $\mathcal{I}$ (Semantic Indexing) 추가, $\mathcal{R}(\text{chunk})$, $\mathcal{R}_{\text{edge}}$ 제거 실험

- $\mathcal{I}$: 저품질 Entity 설명이 오히려 검색 품질을 저하 (가장 큰 $\text{acc}\downarrow$, but $\text{err}\downarrow$)
- $\mathcal{R}_{\text{chunk}}$: 원본 접근하지 못하면서 정보 손실 발생
- $\mathcal{R}_{\text{edge}}$: 관계 정보 손실에 따른 성능 하락

Table 2: Ablation study results comparing accuracy (acc , $\uparrow$) and error rate (err , $\downarrow$) (%) across architectural variants: baseline MiniRAG versus variants with (i) semantic-driven indexing replacement ($-\mathcal{I}$), (ii) edge information removal ($-\mathcal{R}_{\text{edge}}$), and (iii) chunk nodes removal ($-\mathcal{R}_{\text{chunk}}$). Results validate SLM limitations and the effectiveness of query-guided reasoning path components.

LiHuaWorld	MiniRAG		$-\mathcal{I}$		$-\mathcal{R}_{\text{chunk}}$		$-\mathcal{R}_{\text{edge}}$	
	$\text{acc}\uparrow$	$\text{err}\downarrow$	$\text{acc}\uparrow$	$\text{err}\downarrow$	$\text{acc}\uparrow$	$\text{err}\downarrow$	$\text{acc}\uparrow$	$\text{err}\downarrow$
Phi-3.5-mini-instruct	53.29%	23.35%	26.02%	19.12%	48.90%	17.40%	50.47%	15.36%
GLM-Edge-1.5B-Chat	52.51%	25.71%	25.08%	31.50%	46.24%	16.77%	47.81%	20.53%
Qwen2.5-3B-Instruct	48.75%	26.02%	24.14%	15.67%	40.91%	16.14%	48.43%	18.65%
MiniCPM3-4B	51.25%	21.79%	26.18%	15.52%	46.39%	15.83%	48.59%	19.44%

Evaluation: Case Study

Table 3: Case study comparing LightRAG and MiniRAG on a complex restaurant identification query, demonstrating how query-guided reasoning path discovery effectively addresses small language model (SLM) limitations in multi-constraint information retrieval tasks.

Query: What is the name of the Italian restaurant where Wolfgang and Li Hua are having dinner to celebrate Wolfgang's promotion?

Ground-Truth: Venedia Grancaffè

LightRAG Response:

Wolfgang and Li Hua are planning to have dinner at **a pasta joint** to celebrate Wolfgang's promotion...The specific name of the restaurant wasn't listed in the provided information...

MiniRAG Response:

The name of the Italian restaurant where Wolfgang and Li Hua are having dinner to celebrate Wolfgang's promotion is Venedia Grancaffè. Wolfgang mentioned it in a reminder about the dinner planned for 6 PM. They were excited to celebrate his promotion with a feast, with both looking forward to enjoying some delicious food together....

Conclusion & Limit

- **핵심 요약:** SLM RAG 문제 정의 및 SLM Graph RAG 시스템 제안
- **핵심 기여:** Chunk + Entity 이중 노드 구조의 Graph 및 경량 검색 메커니즘 설계
- **주요 성과:** **13.5%p** 향상된 Accuracy와 약 **25%**의 Storage 사용량을 달성
- **미래 가치:** On-device 환경에서 개인화된 데이터 RAG 활용 가능
- **한계점 :**
 - 평가 데이터셋 제한적
 - Graph 품질 의존적
 - **Chunk Size, Top-k** 등에 대한 **sensitivity analysis** 부재

<https://github.com/HKUDS/MiniRAG>